

- We know that $A \cdot 0 = 0$. $\therefore 0 \in S$.
- Let $x, y \in S$. $\therefore Ax = 0$ and $Ay = 0$. $Ax + Ay = 0 + 0$
 $\Rightarrow Ax + Ay = 0$ (Algebraic property iii)
 $\Rightarrow A(x+y) = 0$ (Theorem 5(a) from Chapter 1)

$\therefore x+y \in S$.

- Let $x \in S$ and $c \in \mathbb{R}$. $\therefore Ax = 0 \Rightarrow c(Ax) = c \cdot 0 \Rightarrow c(Ax) = 0$
 $\Rightarrow A(cx) = 0$ (Theorem 5(b) from Chapter 1)

$\therefore cx \in S$.

$\therefore S$ is a subspace of $\mathbb{R}^n$. The given statement is True. $\square$

d) The columns of an invertible $n \times n$ matrix form a basis for $\mathbb{R}^n$.

Solution: Let, A be an invertible $n \times n$ matrix. From the Invertible Matrix Theorem and by the equivalence of the statements (a), (e) and (h), we know that the columns of A form a linearly independent set and the columns of A span $\mathbb{R}^n$. $\therefore$ By defn, the columns of A form a basis for $\mathbb{R}^n$.

$\therefore$ The given statement is True. $\square$

e) Row operations do not affect linear dependence relations among the columns of a matrix.

Solution: The given statement is True. One can see this from Example 7 and Example 8.

22) a) A subset H of $\mathbb{R}^n$ is a subspace if the zero vector is in H .

Solution: By the defn. of subspace, the given statement is False.

b) Given vectors $v_1, \dots, v_p$ in $\mathbb{R}^n$, the set of all linear combinations of these vectors is a subspace of $\mathbb{R}^n$.

Solution: By Example 3, the given statement is True.

c) The null space of an $m \times n$ matrix is a subspace of $\mathbb{R}^n$.

Solution: By Theorem 12, the given statement is True.